UNIVERSITY OF OTAGO EXAMINATIONS 2021

COMPUTER SCIENCE

COSC343

ARTIFICIAL INTELLIGENCE

Semester One

(TIME ALLOWED: THREE HOURS)

This examination comprises 7 pages.

Candidates should answer questions as follows:

Candidates should answer **All** questions.

Questions are worth 12 marks and sub marks are shown thus: (2)

The total number of marks for this exam is 60.

The following material is provided:

Nil.

Use of calculators:

No restriction on the model of calculator that may be used, but no device with communication capability shall be accepted as a calculator. Calculators are subject to inspection by the examiners.

Candidates are permitted copies of:

Nil.

TURN OVER

1. State-space search

(a) Contrast *breadth-first search* (BFS) to *depth-first search* (DFS) in terms of memory usage and optimality guarantees (no need for numerical complexity analysis, just a description of the differences). (2)

(b) What's a good compromise between BFS and DFS, and how does it work? (4)

(c) Recall the application of A* search for solving the 8-puzzle problem from the lectures and tutorials. There were two choices of heuristic: misplaced tile count and the total manhattan distance of each misplaced tile from its goal location. Given that both are admissible and consistent, does it matter which is used in A* in terms of guarantees of finding the optimal solution and the speed of the search? Explain your answer. (3)

(d) Is the *minimax* algorithm with *alpha-beta* pruning a DFS or BFS? (1)

(e) What's the purpose of *alpha-beta* pruning in the *minimax* search, and under what conditions does it work best? (2)

2. Learning

(a) What's the difference between a *deterministic* and *stochastic* environment? Give an example of each. (2)

(b) Why do we split data into *training* and *testing* sets in *supervised* learning? What is a *validation* set and how is it used during training? (3)

(c) Given an update rule $\mathbf{w} := \mathbf{w} + \alpha\Delta\mathbf{w}$ for the *steepest gradients descent* optimisation, where $\mathbf{w}$ is a vector of model parameters and $\Delta\mathbf{w}$ is the negative gradient of some cost function with respect to parameters, what is the role of the α in this equation? What is the general guideline for setting the value of α and how does this choice affect the dynamics of the optimisation? (3)

(d) What is a gradient, and why does $\Delta\mathbf{w}$ in the above update rule need to be a negative gradient? General explanation is sufficient here (no need to provide mathematical formulas). (2)

(e) Suppose we are training a model on a supervised regression problem using the *mean squared error* loss. Does reaching the global minimum of the loss surface during training imply consistency on all training data? What about the reverse – does consistency imply reaching the global minimum during training? Explain your answers. (2)

3. Classification

Consider the following version of the Iris dataset:

sepal length (SL)	sepal width (SW)	petal length (PL)	Iris type (IT)
short	big	short	Setosa
long	medium	long	Virginica
long	big	long	Versicolor
short	medium	short	Setosa
long	big	medium	Versicolor
medium	medium	short	Setosa
⋮	⋮	⋮	⋮

For the purpose of this exercise, the length and width measurements are expressed as three possible values from the set {short, medium, long} and {small, medium, big} respectively. The objective is to create an agent that classifies whether a plant with a set of those three measurements is a Setosa, Virginica or Versicolor.

(a) Describe the computation for the Naïve Bayes Classification of this problem. You don't need to evaluate the actual probabilities, just give the formula in terms of the probabilities of interest and explain how the decision is made based on those formulas. Use SL, SW, PL, IT for your random variables. For reference, Bayes' rule is:

$$P(\text{effect}|\text{causes}) = \frac{P(\text{causes}|\text{effect})P(\text{effect})}{P(\text{causes})} \tag{4}$$

(b) Describe how a decision tree can be built and would work to perform this classification. No need for numerical evaluation and in this case no need for formulas – just a generic description of the process and the computation involved is sufficient. (4)

(c) How would you format the data in order to train a perceptron on it? Reproduce the table above with your proposed data format. (2)

(d) Specify the architecture of the perceptron suitable for this problem, including the number of weights and biases. If you'd like, you can just draw a diagram labelling all inputs, outputs and parameters involved. (2)

4. **Neural networks**

(a) What's the difference between *perceptron* and *multi-layer perceptron (MLP)* in terms of architecture, capabilities and training? (4)

(b) What's the difference between an *MLP* and a *convolutional neural network (CNN)* in terms of their architecture and capabilities? (4)

(c) What's the difference between a *feed forward* and a *recurrent neural network* in terms of their architecture and capabilities? (2)

(d) When is it appropriate to use the *mean squared error* and when the *cross-entropy* loss functions when training an MLP? (1)

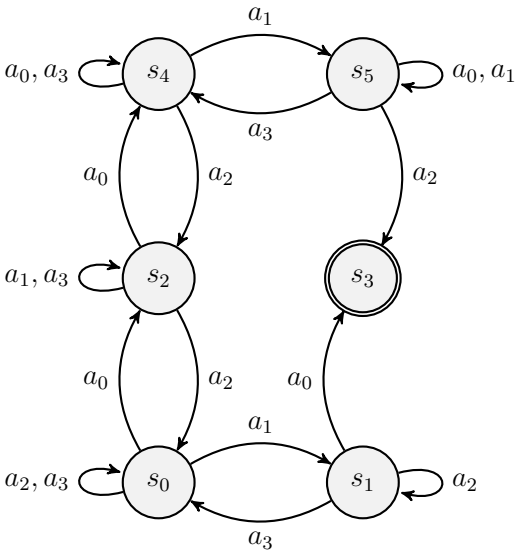
(e) What's one-hot encoding? (1)

5. Sequential decisions

Consider an environment of 6 discrete states $s_0, \ldots, s_5$, four actions $a_0, \ldots, a_3$, and reward function:

$$R(s) = \begin{cases} 1 & \text{when } s = s_3 \\ -0.04 & \text{otherwise.} \end{cases}$$

The environment is governed by the deterministic transition model and the agent by the policy given respectively in the following state machine and Q-table:



State machine

	a_0	a_1	a_2	a_3
s_0	-0.01	0.90	0.03	0.02
s_1	0.99	0.04	0.02	0.01
s_2	-0.02	0.02	0.43	-0.00
s_3	1.00	1.00	1.00	1.00
s_4	-0.01	0.00	0.01	-0.01
s_5	0.00	0.15	0.09	0.02

Q-table

(a) What is an *optimal policy*? (1)

(b) Assuming initial state s_0 and terminal state s_3, what is the sequence of actions of the agent according to the policy given in the Q-table above. Assuming no discount on the future reward, what's the reward for this sequence of actions? Is this policy optimal for this starting state? If not, what changes to the table would need to be made to make it optimal and what would the reward be then? (4)

(c) Assuming initial state s_4 and terminal state s_3, what is the sequence of actions of the agent according to the policy given in the Q-table above. Assuming no discount on the future reward, what's the reward for this sequence of actions? Is this policy optimal for this starting state? If not, what changes to the table would need to be made to make it optimal and what would the reward be then? (4)

(d) If the environment was different such that

$$R(s) = \begin{cases} 1 & \text{when } s = s_3 \\ 0 & \text{otherwise,} \end{cases}$$

what could be done during training in order to encourage the agent to traverse the least number of states from its initial state to the terminal one? (1)

(e) What is the general technique in reinforcement learning that gives the agent a chance to discover the best solution given the existence of several different ways of solving a problem? Explain how this technique works. (2)

END